Q8). **Explain the TCP segment header with the help of a diagram. Explain the process of TCP connection establishment using a three-way handshake mechanism. Also, explain the TCP flow control with the help of an example.**

ans. The Transmission Control Protocol (TCP) is a reliable, connection-oriented protocol used for transmitting data over networks. It ensures that data is delivered accurately, in order, and without loss. To achieve this, TCP uses a well-defined segment structure, connection establishment process, and flow control mechanism.

TCP Segment Header

A TCP segment consists of a header and data. The header contains important control information required for reliable communication.

Main Fields of TCP Header

1. Source Port and Destination Port: Identify sender and receiver applications
2. Sequence Number: Indicates position of data in the stream
3. Acknowledgment Number: Confirms receipt of data
4. Header Length: Size of the TCP header
5. Flags: Control bits such as SYN, ACK, FIN
6. Window Size: Used for flow control
7. Checksum: Error detection
8. Urgent Pointer: Points to urgent data

Diagram of TCP Header

Source Port	Destination Port	
Sequence Number		
Acknowledgment Number		
HLEN	Flags	Window Size
Checksum	Urgent Pointer	
Options (if any)		

TCP Connection Establishment (Three-Way Handshake)

Before data transmission begins, TCP establishes a connection using a three-step process known as the three-way handshake.

Step 1: SYN

The client sends a segment with the SYN (synchronize) flag set and an initial sequence number.

Step 2: SYN-ACK

The server responds with a segment that has both SYN and ACK flags set. It acknowledges the client's request and sends its own sequence number.

Step 3: ACK

The client sends an acknowledgment (ACK) back to the server, completing the connection.

Client ⟶ SYN ⟶ Server

Client ⟵ SYN + ACK ⟵ Server

Client ⟶ ACK ⟶ Server

TCP Flow Control

Flow control ensures that the sender does not overwhelm the receiver with too much data.

TCP uses a sliding window mechanism for this purpose. The receiver specifies a window size, which indicates how much data it can accept at a time.

Working

1. Sender sends data within the window limit
2. Receiver sends acknowledgment along with updated window size
3. Window "slides" forward as data is acknowledged

Example

If the window size is 3, the sender can send 3 segments without waiting for acknowledgment. After receiving ACK, it sends the next set of segments.

Thus, TCP ensures reliable communication through structured headers, a secure connection establishment process, and efficient flow control mechanisms.

Q9). a) What are the security considerations of WWW? Discuss the use of SSL and S-HTTP for web security.

b) Explain the Diffie-Hellman algorithm. Also, explain an example of its use.

c) Explain how digital signatures are used in the context of network security. What are the uses of digital signatures?

ans. The World Wide Web (WWW) is widely used for communication and information exchange, but it is also vulnerable to various security threats. Therefore, ensuring security in web communication is extremely important.

Security Considerations of WWW

Some common security issues include:

1. Confidentiality: Preventing unauthorized access to data
2. Integrity: Ensuring data is not altered during transmission
3. Authentication: Verifying identity of users and servers
4. Availability: Ensuring services remain accessible

Web applications are often exposed to attacks such as eavesdropping, data tampering, and impersonation.

SSL (Secure Sockets Layer)

SSL is a protocol designed to provide secure communication over a network. It works by encrypting the data exchanged between client and server.

Features of SSL

Encryption of data

1. Server authentication
2. Data integrity
3.

SSL uses a combination of symmetric and asymmetric encryption to ensure security.

HTTPS (HyperText Transfer Protocol Secure)

HTTPS is the secure version of HTTP, which uses SSL/TLS to encrypt communication.

Working

When a user accesses a website using HTTPS, a secure connection is

1. established

Data is encrypted before transmission

2. Ensures privacy and security
3.

Thus, SSL and HTTPS play a crucial role in protecting web communication from various security threats.

b) Diffie-Hellman Algorithm

The Diffie-Hellman algorithm is a key exchange mechanism that allows two parties to securely share a secret key over an insecure channel.

Working Principle

1. Two users agree on:
   - A large prime number p
   - A base g
2. Each user selects a private key:
   - User A chooses a
   - User B chooses b
3. They compute public values:
   - A = g^a mod p
   - B = g^b mod p
4. Exchange public values
5. Compute shared key:
   - User A: K = B^a mod p
   - User B: K = A^b mod p

Both get the same secret key

Example

Let:

$p = 23, \quad g = 5$

A chooses $a = 6$, B chooses $b = 15$

Then:

$A = 5^6 \bmod 23 = 8$

$B = 5^{15} \bmod 23 = 19$

Shared key:

$K = 19^6 \bmod 23 = 2$

Use of Diffie-Hellman

- Secure key exchange in HTTPS
- VPN connections
- Secure messaging systems

c) Digital Signatures in Network Security

A digital signature is a cryptographic technique used to verify the authenticity and integrity of a message or document.

Working of Digital Signature

1. Sender creates a message
2. A hash function is applied to generate a message digest
3. The digest is encrypted using the sender's private key
4. This encrypted digest is the digital signature

At the receiver:

- The signature is decrypted using the sender's public key
- The receiver also computes the hash
- If both match, the message is verified

Message ⟶ Hash ⟶ Encrypt (Private Key) ⟶ Signature

Receiver ⟶ Decrypt (Public Key) ⟶ Compare Hash

Uses of Digital Signatures

- Authentication of sender
- Data integrity verification
- Non-repudiation (sender cannot deny sending message)
- Used in: Email security, Online transactions, Software distribution